

EventForce Management System

A Salesforce CRM for event planners: bookings, venues, vendors, approvals and feedback in one place.

TEAM ID	PLATFORM	ORG TYPE	AUTOMATION	TEAM SIZE
SWTID-2026-3509	Salesforce	Developer Edition	Flow, Approval, Apex	4 members

OUR TEAM

Naveen P	TEAM LEADER	NM ID: 2F9886E9DAD1704907D4F8C339C7228D Register No: 821923243034
Saravanan K	TEAM MEMBER	NM ID: F18CF35068E4C2B82EE447E9431781E7 Register No: 821923243046
Roopan G	TEAM MEMBER	NM ID: F85EB53C24A0D469DCC4BCD5ABF04272 Register No: 821923243043
Mukil Doss S	TEAM MEMBER	NM ID: 0F27DB0131DA011A5144D5726477679E Register No: 821923243033

Contents

01 Project Overview 3

02 Phase 1: Requirement Analysis and Planning 4

03 Phase 2: Backend Development and Configuration 8

04 Phase 3: UI/UX Development and Customization 18

05 Phase 4: Data Migration, Testing and Security 22

06 Plan and Schedule 29

07 Strengths and Limits 31

08 Conclusion and Roadmap 32

A Appendix 33

The project at a glance

4 team members	6 custom objects	1 Record-Triggered Flow	1 Approval Process
5 Apex classes and triggers	4 profiles and 4 roles	2 reports, 1 dashboard	1 permission set, 1 sharing rule

01 Project Overview

What EventForce is and why we built it

Event planning companies handle many things at the same time: client requests, event scheduling, venue reservations, vendor assignments, cancellations and feedback. When these live in notebooks, chats and spreadsheets, bookings clash, vendors are missed and clients are not kept informed. The EventForce Management System brings all of this into one Salesforce CRM, where every client, event, venue, vendor and feedback record sits in a single connected data model.

The application is built in a Salesforce Developer Edition org. Custom objects hold the data, a validation rule keeps contact data clean, a Flow reminds clients before their event, an Approval Process controls cancellations, Apex stops double bookings and keeps venue status current, and reports with a dashboard give the team a live view of upcoming events. Access is controlled with profiles, roles, a permission set, organization-wide defaults and a sharing rule.

Purpose and objectives

- Centralize client, event, venue, vendor and feedback information in one connected CRM.
- Streamline event booking and venue reservation, and prevent a venue from being booked twice on the same date.
- Update venue availability automatically when an event is confirmed or cancelled.
- Remind clients three days before their event without manual follow-up.
- Route cancellation requests through a manager approval with email alerts.
- Give each type of user only the access that their job needs.
- Show upcoming events through reports and a dashboard for better decisions.

Technology used

Layer	Salesforce feature used
Platform	Salesforce CRM, Developer Edition org
Data model	Custom objects, Lookup and Master-Detail relationships, formula field, lookup filter
Business logic	Validation rule, Record-Triggered Flow (scheduled path), Approval Process, Email Alerts, Field Updates
Code	Apex class, Apex triggers, Batch Apex and Schedulable Apex (API version 67)
User interface	Custom tabs, Lightning App (Event Planner), Reports and Dashboards
Security	Profiles, Roles, Users, Permission Set, Organization-Wide Defaults, Sharing Rule
Data loading	Salesforce Data Import Wizard

02 Phase 1: Requirement Analysis and Planning

Problems, scope, stakeholders and roadmap

Problem statements

ID	Who	Is trying to ... but ...	Feels
PS-1	Event Operations Manager	manage event planning efficiently, but client, event, venue and vendor information is hard to manage centrally.	Frustrated
PS-2	Event Coordinator	create and manage bookings, but there is no central platform and many steps are manual.	Overwhelmed
PS-3	Venue Manager	manage venue availability, but the same venue may be assigned to several events.	Uncertain
PS-4	Vendor Manager	coordinate vendors for events, but vendor details and assignments are managed separately.	Overwhelmed
PS-5	Event Coordinator	send reminders on time, but reminders are handled manually.	Stressed
PS-6	Event Administrator	manage cancellation requests, but they need a controlled approval and status updates.	Frustrated
PS-7	Event Coordinator	manage client feedback, but it must stay linked to both the event and the client.	Concerned
PS-8	Event Operations Manager	monitor operations and performance, but data is not organized into reports and dashboards.	Less confident

Ideas considered and final choice

Idea	Approach	Key limitation
Manual registers	Write event, client, venue and vendor details by hand	Time-consuming and hard to maintain
Spreadsheet tracking	Track bookings in spreadsheets	Limited automation and no real-time coordination
Standalone portal	Build a dedicated event portal	High development and maintenance effort
Salesforce CRM (chosen)	Centralize event operations on Salesforce	Needs proper configuration, which this project provides

Decision. Salesforce CRM was chosen for automation (reminders, approvals, status updates), scalability (more events, users and records) and real-time visibility through reports and dashboards.

Functional scope

- **Data model:** six custom objects: Event, Client, Vendor, Venue, Feedback and Event Vendor (junction).
- **Relationships:** Lookup, Master-Detail and a many-to-many link between Event and Vendor.
- **Validation:** an email validation rule and a formula field for event budget.
- **Automation:** a Flow for client reminders and an Approval Process for cancellations.
- **Apex:** venue status updates, double-booking prevention and a scheduled batch job.

- **Security:** profiles, roles, a permission set, organization-wide defaults and a sharing rule.
- **Insights:** reports and a dashboard on upcoming events.

Stakeholders

Stakeholder	What they do in the system
Client	Requests event services, gives feedback and follows event updates.
Event Coordinator	Schedules events, talks to clients, raises cancellation requests and follows up on reminders.
Vendor Manager	Handles vendors for services such as catering, decor and logistics.
Venue Manager	Manages venue availability and reservations.
Event Admin	Oversees the system, users, access and data.

Customer journey

#	Stage	What happens in Salesforce
1	Enquiry and client registration	Client requirements and contact details are captured.
2	Planning and booking	Event type, date, budget and status are created and managed.
3	Venue selection	Availability is checked and the venue is reserved without double booking.
4	Vendor coordination	Vendors and services are assigned to the event through Event Vendor.
5	Confirmation and communication	Confirmed events trigger the reminder email to the client.
6	Execution and monitoring	Progress is followed through records, reports and the dashboard.
7	Cancellation and approval	Requests go through the approval process with status updates and alerts.
8	Feedback and closure	Feedback is captured and past events are marked Completed.

Solution requirements

Functional requirements

No.	Requirement	No.	Requirement
FR-1	Event management	FR-7	Automated event reminders
FR-2	Client management	FR-8	Event cancellation approval
FR-3	Venue management	FR-9	Feedback management
FR-4	Vendor management	FR-10	Reports and dashboards
FR-5	Event and vendor assignment	FR-11	Event status automation
FR-6	Venue availability and double-booking prevention	FR-12	Data migration through Data Import Wizard

Non-functional requirements

Quality	Description
Usability	Lightning App for easy event management and navigation.
Security	Profile, role and permission based access control.
Reliability	Accurate automation, approvals and venue booking validation.

Quality	Description
Performance	Efficient processing of event and operational data.
Availability	Cloud-based Salesforce access for authorized users.
Scalability	Supports more events, clients, vendors, venues and users.
Maintainability	Flows, Apex, validation rules and security can be monitored and updated.
Data integrity	Validation rules and controlled data migration keep records accurate.

Solution architecture and data flow

The diagram shows how users reach the system and how each layer builds on the one below it. Supporting flows run alongside the main flow: **Cancellation** (request, approval, status update, notification), **Venue booking validation** (event saved, Apex trigger checks venue and date, error or status update) and **Event completion** (date passed, scheduled batch, status set to Completed).

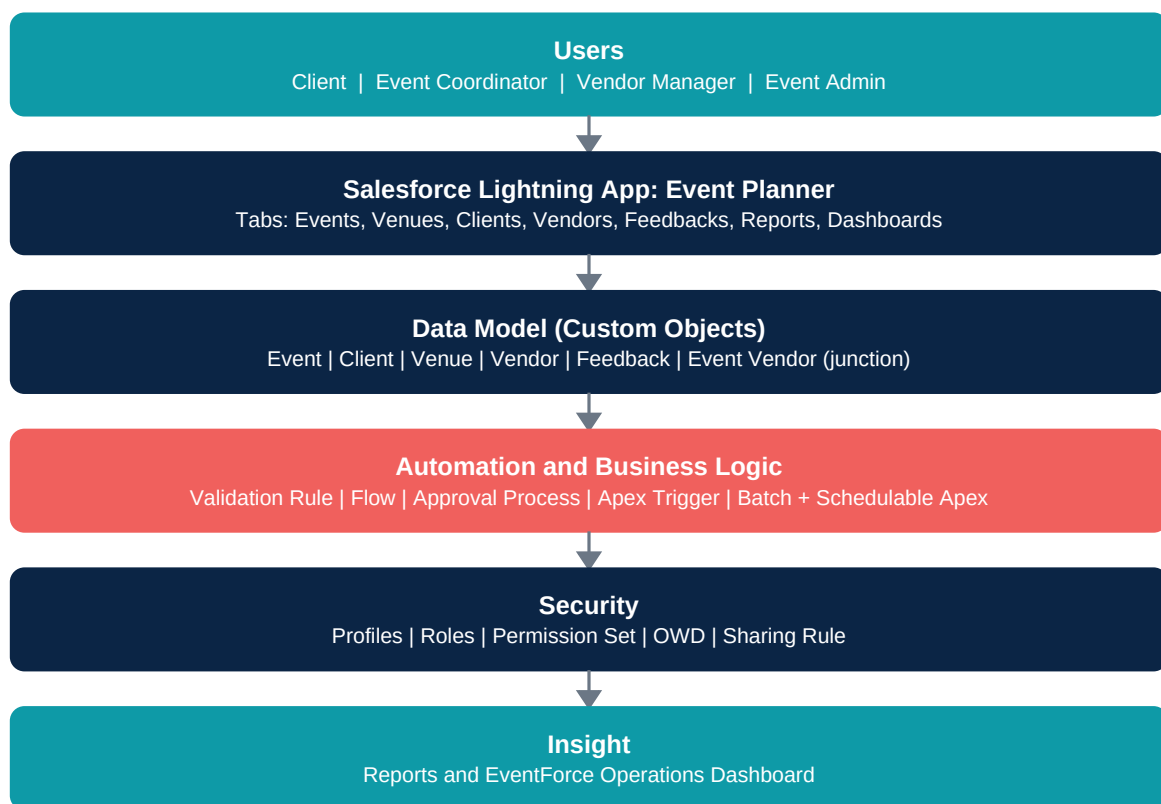


Figure A. Layered solution architecture of EventForce.

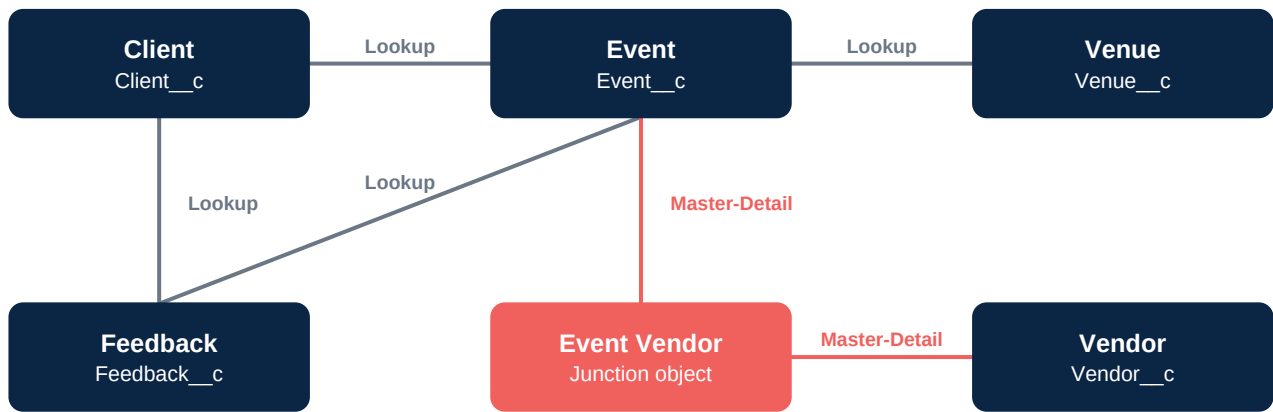


Figure B. Data model with Lookup and Master-Detail relationships.

Problem and solution fit

Problem	EventForce solution	Benefit
Information is scattered	Custom objects for Event, Client, Venue, Vendor, Feedback, Event Vendor	Central, structured data
Venue double booking	PreventDoubleBooking trigger and VenueStatusHelper	Venue schedule stays accurate
Vendor tracking is hard	Event Vendor junction object	Many-to-many assignment
Late client reminders	3-day reminder Flow	Reminders without manual work
Uncontrolled cancellations	Event_Cancellation_Process	Manager approval and status updates
Feedback not linked	Feedback with Event and Client lookups	Organized feedback
No visibility	Reports and Operations Dashboard	Quick view of upcoming events
Different access needs	Profiles, roles, permission set, OWD, sharing rule	Secure, controlled access

Execution roadmap

Step	Work item	Phase
1	Create the Developer org, the six custom objects, fields and relationships	Phase 2
2	Add the Client email validation rule	Phase 2
3	Build the Event cancellation Approval Process and email templates	Phase 2
4	Build the 3-day Client Reminder Flow	Phase 2
5	Write Apex classes and triggers for venue status, double booking and the batch job	Phase 2
6	Create tabs, the Event Planner app, reports and dashboard	Phase 3
7	Load data, then set up profiles, roles, users, permission set and sharing settings	Phase 4
8	Test the features and record the results	Phase 4

03 Phase 2: Backend Development and Configuration

Org setup, data model, automation and Apex

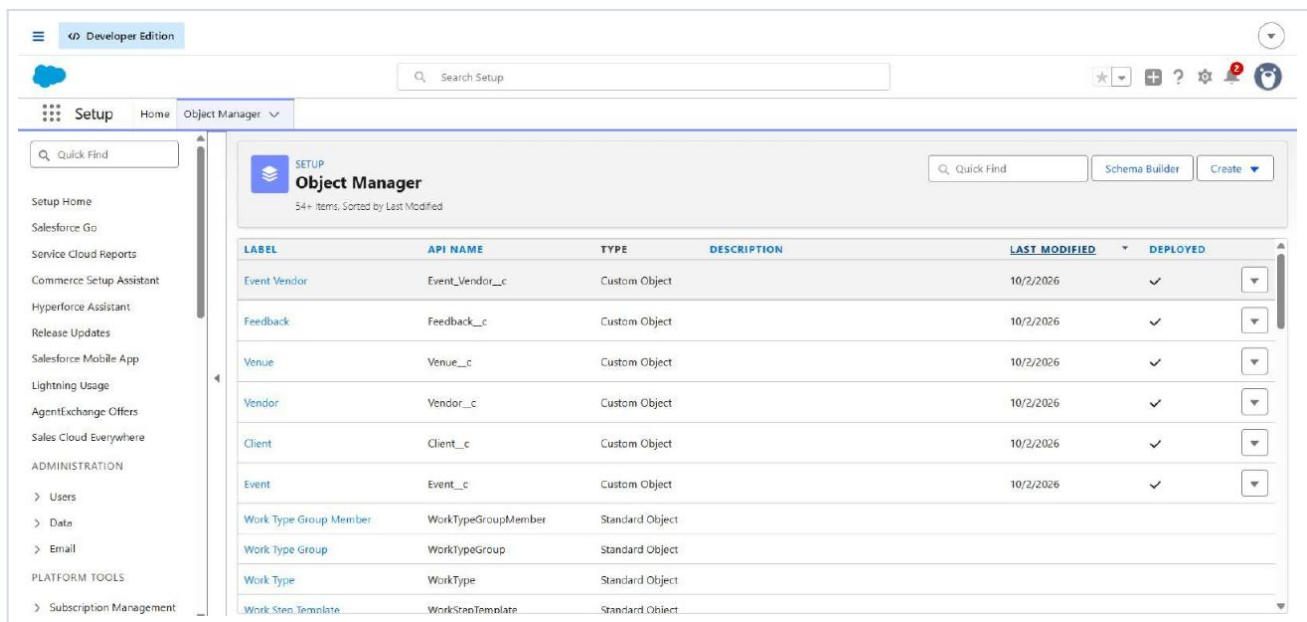
This phase built the working engine of EventForce. Declarative tools (validation rule, Flow, Approval Process, formula field) handle the routine rules, and Apex handles the logic that needs code, such as venue availability and double-booking prevention.

Milestone 1: Salesforce Developer org

- Open the Salesforce developer sign-up page and fill in first name, last name, email, role (Developer), company (college name), country (India), postal code and a unique username in the form *name@organization.com*.
- Open the verification email (it can take 5 to 10 minutes), click **Verify Account**, set a password and answer the security question.
- You are then taken to Salesforce Setup. All objects, automation and code in this project were built in this org.

Milestone 2: Custom objects

Five main objects were created from **Setup > Object Manager > Create > Custom Object: Event, Client, Vendor, Venue and Feedback**. A sixth object, **Event Vendor**, is a junction object. For each object, Allow Reports, Allow Search and the Notes and Attachments related list were enabled, and the object was Deployed. Event also has Allow Activities and Track Field History. Feedback uses an Auto Number record name with the display format **F-{0000}** starting at 0.



The screenshot shows the Salesforce Setup interface in the 'Developer Edition'. The 'Object Manager' tab is selected, displaying a list of 54 items sorted by 'Last Modified'. The list includes several custom objects: Event_Vendor__c, Feedback__c, Venue__c, Vendor__c, Client__c, and Event__c, all created on 10/2/2026 and marked as 'Deployed'. It also lists standard objects like Work Type Group Member, Work Type Group, Work Type, and Work Stan Template.

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Event_Vendor	Event_Vendor__c	Custom Object		10/2/2026	✓
Feedback	Feedback__c	Custom Object		10/2/2026	✓
Venue	Venue__c	Custom Object		10/2/2026	✓
Vendor	Vendor__c	Custom Object		10/2/2026	✓
Client	Client__c	Custom Object		10/2/2026	✓
Event	Event__c	Custom Object		10/2/2026	✓
Work Type Group Member	WorkTypeGroupMember	Standard Object			
Work Type Group	WorkTypeGroup	Standard Object			
Work Type	WorkType	Standard Object			
Work Stan Template	WorkStanTemplate	Standard Object			

Figure 1. Object Manager showing the custom objects Event Vendor, Feedback, Venue, Vendor, Client and Event.

Milestone 3: Fields and relationships

Object	Key fields
Event	Event Name (Text 80), Client (Lookup), Venue (Lookup), Event Date (Date), Event Type (Picklist), Event Status (Picklist), Event Budget (Formula, Currency)
Client	Client Name, Phone, Email, Address (Text Area 255), City (Picklist), Country (Picklist)
Vendor	Vendor Name, Phone, Email, Service Type (Multi-select Picklist), Status (Picklist)

Object	Key fields
Venue	Venue Name, Address (Text Area 255), Location (URL), Capacity (Number), Availability Status (Picklist)
Feedback	Feedback (Auto Number), Client (Lookup), Event (Lookup), Rating (Picklist), Comments (Long Text Area)
Event Vendor	Event (Master-Detail), Vendor (Master-Detail)

Picklist values

Field	Values
Event Type	Wedding, Corporate, Birthday, Anniversary, Festival, Concert, Other
Event Status	Planned, Confirmed, Completed, Pending Cancellation, Canceled, Rejected
Vendor Service Type	Catering, Decor, Photography, Videography, Lighting, Stage Setup, Makeup Artist, DJ/Music, Transportation, Hosting/Anchor
Vendor Status	Available, Booked, Cancelled
Venue Availability Status	Available, Reserved, Not Available
Feedback Rating	1, 2, 3, 4, 5
Client Country	India, Philippines, Saudi Arabia, USA, Switzerland, Thailand, Maldives, Dubai, Sri Lanka, Bangkok, Singapore, Myanmar, France, Nepal

Event Budget formula field

Event Budget is a Formula field (return type Currency) that sets the budget from the event type:

Formula

```
CASE( TEXT(Event_Type__c),
  "Wedding", 50000, "Corporate", 30000, "Birthday", 10000,
  "Anniversary", 20000, "Festival", 60000, "Concert", 40000,
  "Other", 15000, 0 )
```

Relationship	Type	Purpose
Event to Client	Lookup	Each event belongs to a client
Event to Venue	Lookup	Each event is held at a venue
Feedback to Event	Lookup	Feedback is given for an event
Feedback to Client	Lookup	Feedback comes from a client
Event Vendor to Event	Master-Detail	Junction side one
Event Vendor to Vendor	Master-Detail	Junction side two; gives the many-to-many link

The screenshot shows the Salesforce Setup interface in Developer Edition. The left sidebar contains navigation links for Setup Home, Salesforce Go, Service Cloud Reports, Commerce Setup Assistant, Hyperforce Assistant, Release Updates, Salesforce Mobile App, Lightning Usage, AgentExchange Offers, Sales Cloud Everywhere, ADMINISTRATION (Users, Data, Email), and PLATFORM TOOLS (Subscription Management). The main content area is titled 'SETUP > OBJECT MANAGER Event'. It features a 'Details' sidebar with links to Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, and List View Button Layout. The 'Fields & Relationships' section is active, showing 10 items sorted by Field Label. The table below lists these fields with their labels, names, data types, controlling fields, and indexed status.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Client	Client__c	Lookup(Client)		✓
Created By	CreatedById	Lookup(User)		
Event Budget	Event_Budget__c	Formula (Currency)		
Event Date	Event_Date__c	Date		
Event Name	Name	Text(80)		✓
Event Status	Event_Status__c	Picklist		
Event Type	Event_Type__c	Picklist		

Figure 2. Event object: fields and relationships (Client lookup, Event Budget formula, picklists).

The screenshot shows the Salesforce Setup interface in Developer Edition, similar to Figure 2 but for the 'Client' object. The 'Fields & Relationships' section is active, showing 9 items sorted by Field Label. The table below lists these fields with their labels, names, data types, controlling fields, and indexed status.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Address	Address__c	Text Area(255)		
City	City__c	Picklist		
Client Name	Name	Text(80)		✓
Country	Country__c	Picklist		
Created By	CreatedById	Lookup(User)		
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User.Group)		✓

Figure 3. Client object fields.

Developer Edition

Setup Home Object Manager

Quick Find

Setup Home
Salesforce Go
Service Cloud Reports
Commerce Setup Assistant
Hyperforce Assistant
Release Updates
Salesforce Mobile App
Lightning Usage
AgentExchange Offers
Sales Cloud Everywhere
ADMINISTRATION
Users
Data
Email
PLATFORM TOOLS

SETUP > OBJECT MANAGER
Vendor

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout

Fields & Relationships
8 Items, Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedBy	Lookup(User)		
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Phone	Phone__c	Phone		
Service Type	Service_Type__c	Picklist (Multi-Select)		
Status	Status__c	Picklist		
Vendor Name	Name	Text(80)		✓

Figure 4. Vendor object fields.

Developer Edition

Setup Home Object Manager

Quick Find

Setup Home
Salesforce Go
Service Cloud Reports
Commerce Setup Assistant
Hyperforce Assistant
Release Updates
Salesforce Mobile App
Lightning Usage
AgentExchange Offers
Sales Cloud Everywhere
ADMINISTRATION
Users
Data
Email
Subscription Management
PLATFORM TOOLS

SETUP > OBJECT MANAGER
Venue

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout

Fields & Relationships
8 Items, Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Address	Address__c	Text Area(255)		
Availability Status	Availability__c	Picklist		
Capacity	Capacity__c	Number(18, 0)		
Created By	CreatedBy	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Location	Location__c	URL(255)		
Owner	OwnerId	Lookup(User,Group)		✓
Venue Name	Name	Text(80)		✓

Figure 5. Venue object fields.

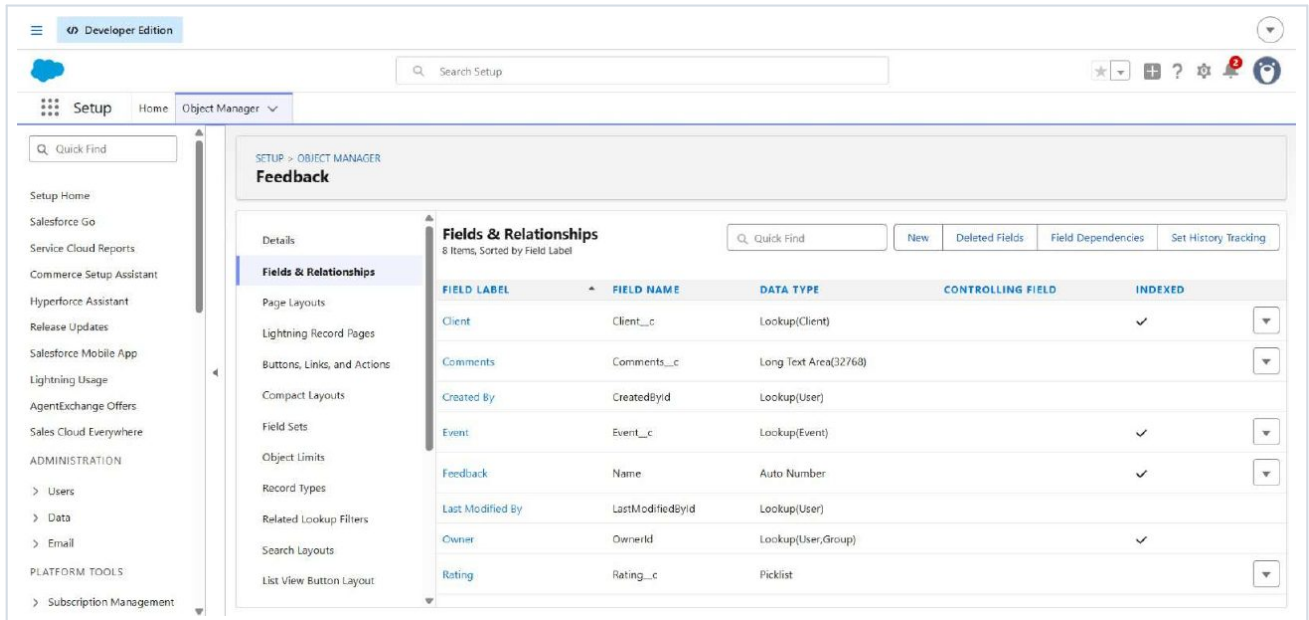


Figure 6. Feedback object: Client and Event lookups with Rating and Comments.

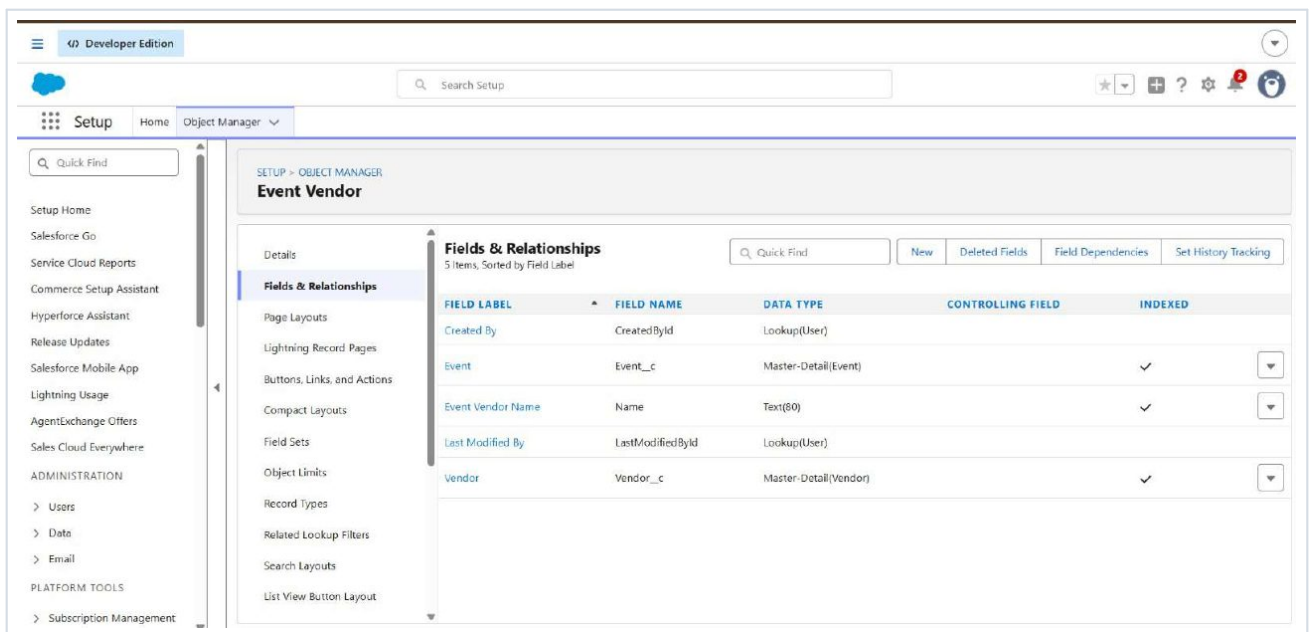


Figure 7. Event Vendor junction object with two Master-Detail relationships.

Lookup filter on Feedback

When a user creates Feedback and picks a Client, the Event lookup should show only that client's events. On the Feedback object, the Event lookup field was edited and a lookup filter was enabled: **Event: Client equals Feedback: Client.**

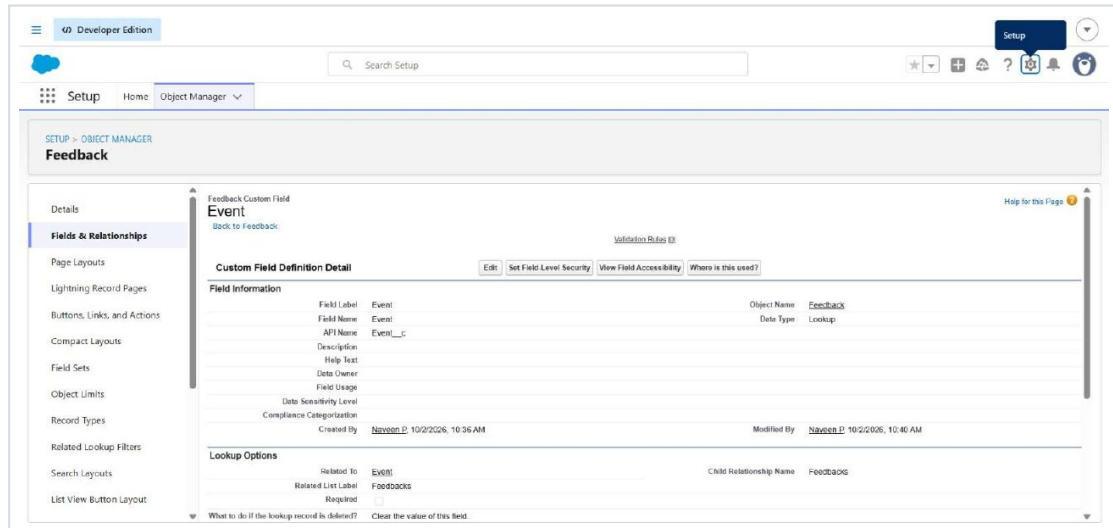


Figure 8. Feedback object: Event lookup field definition (lookup filter configured through Edit).

Milestone 4: Validation rule

To keep contact data clean, the Client object has an active validation rule named **Email_Valid_Address**. It checks the Email field against a pattern and blocks the save when the format is wrong. The error message *Please Enter Valid Email Address* is shown on the Email field.

Error condition formula

```
NOT(REGEX(Email__c, "^[a-zA-Z0-9._]+@[a-zA-Z0-9._]+\.[a-zA-Z]{2,}$"))
```

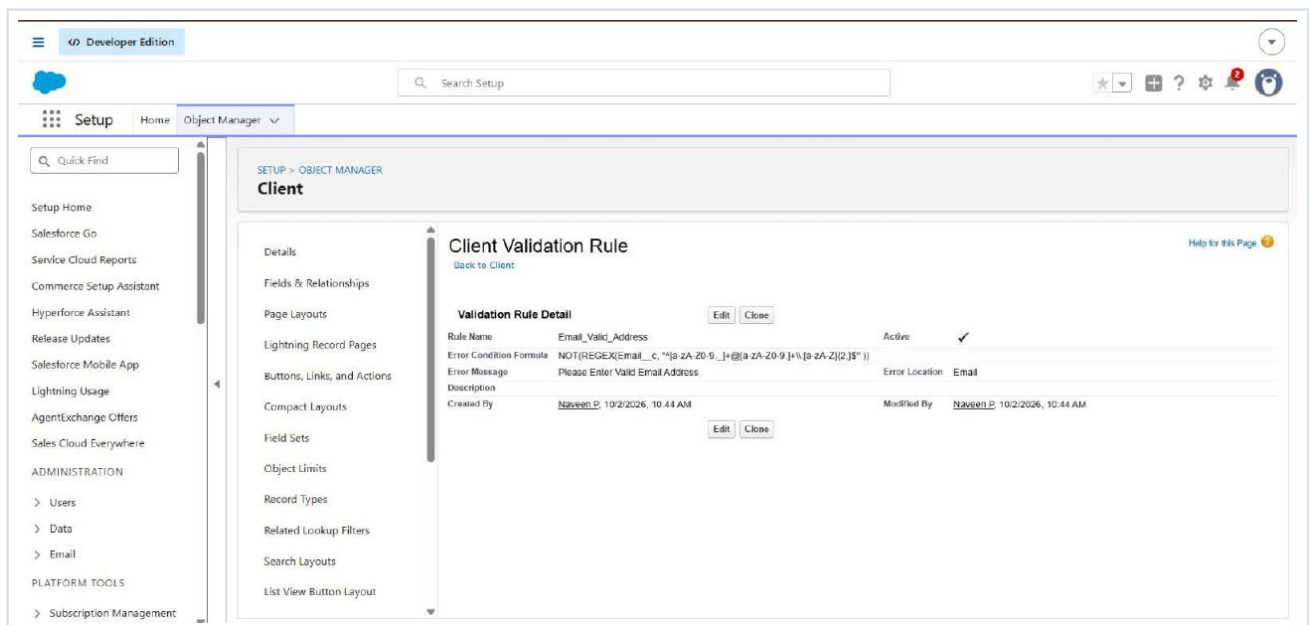


Figure 9. Client validation rule Email_Valid_Address.

Milestone 5: Approval Process for event cancellation

Before the approval process, four classic email templates were created (Setup > Email Templates):

Template	Subject	Used for
Event Cancellation Request Notification	Approval Request: Cancel Event {Event__c.Name}	Approval assignment email
Email to Client After Cancellation	Event Cancellation Notice - {Event__c.Name}	Informs the client

Template	Subject	Used for
Approval Notification to Event Owner	Event Cancellation Approved: {!Event__c.Name}	Sent on approval
Rejection Notification to Event Owner	Event Cancellation Rejected: {!Event__c.Name}	Sent on rejection

The process **Event_Cancellation_Process** runs on the Event object. A record enters the approval when **Event Status equals Pending Cancellation**. The event owner is the initial submitter, the next approver is the manager of the record submitter, and a single step, **Manager Review**, sends the request for approval. The process is activated after configuration.

Stage	Actions configured
Initial submission	Record Lock; Email Alert "Alert - Event Cancellation Request"
Final approval	Field Update "Event Status Update" (status set to Canceled); Email Alert "Alert - Cancellation Approved" to the event owner
Final rejection	Field Update "Event Status To Rejected"; Email Alert "Alert - Cancellation Rejected" to the event owner; record unlocked
Recall	Record Unlock

The screenshot shows the 'Approval Processes' setup page in Salesforce. The process 'Event_Cancellation_Process' is active. Key details include:

- Process Definition Detail:**
 - Process Name: Event_Cancellation_Process
 - Unique Name: Event_Cancellation_Process
 - Description: Event: Event status equals Pending Cancellation
 - Entry Criteria: Event: Event status equals Pending Cancellation
 - Record Editability: Administrator OR Current Approver
 - Approval Assignment Email Template: Event_Cancellation_Request_Notification
 - Initial Submitters: Event Owner
 - Created By: Navleen P. 10/2/2026, 12:03 PM
 - Modified By: Navleen P. 10/2/2026, 12:21 PM
- Initial Submission Actions:**
 - Action: Record Lock (Type: Record Lock, Description: Lock the record from being edited)
 - Action: Email Alert (Type: Email Alert, Description: Alert - Event Cancellation Request)

Figure 10. Approval Process detail: entry criteria, submitters and initial submission actions (process Active).

Milestone 6: Record-Triggered Flow

The Flow **Client Reminder - 3 Days Before** is a Record-Triggered Flow on the Event object. It starts when an Event is created or updated with **Event Status = Confirmed**. It has a **Run Immediately** path and a scheduled path, **3_Days_Before**, with the time source **Event Date** and an offset of 3 days before. The scheduled path runs the Send Email action **Alert_Client_3Day_Reminder**.

- Two text templates hold the message: a subject (*Reminder: Your Event {!\$Record.Name} is in 3 days*) and a body with client name, event name, date, venue and event type.
- The email goes to the Client's email address, with the event owner in CC.
- The Flow is saved and Active.

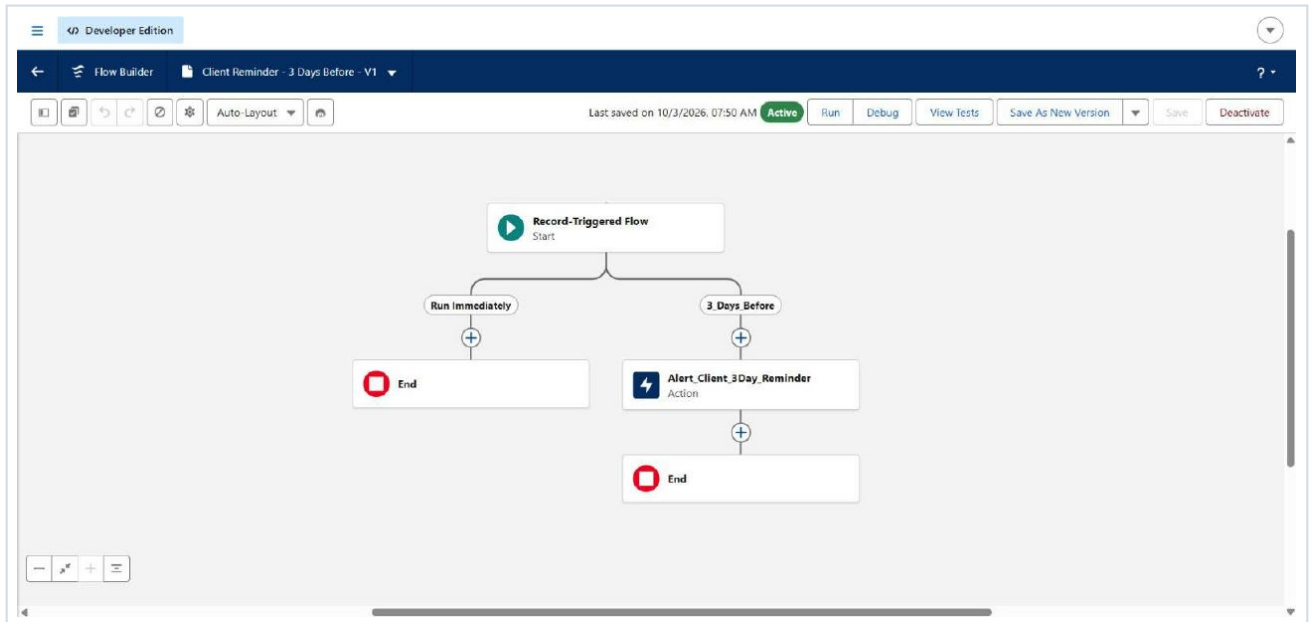


Figure 11. Flow Builder: Client Reminder - 3 Days Before (Active).

Milestone 7: Apex classes and triggers

Apex handles the business rules that need code. All code is written in API version 67 using the Developer Console.

Component	Type	What it does
VenueStatusHelper	Apex class (with sharing)	Sets the venue to Reserved when the event is Confirmed and back to Available when it is Canceled.
EventTrigger13	Trigger on Event (after insert, after update)	Calls VenueStatusHelper whenever an event is saved.
PreventDoubleBooking	Trigger on Event (before insert, before update)	Adds the error "This Venue is already booked on this date." when another event uses the same venue and date.
BatchCompleteEvents	Batch Apex (Database.Batchable)	Finds events dated before today that are not Completed and marks them Completed.
ScheduleCompleteEvents	Schedulable Apex	Runs BatchCompleteEvents with a batch size of 200.

VenueStatusHelper.cls

```

public with sharing class VenueStatusHelper {
    public static void updateVenueStatus(List<Event__c> events){
        List<Venue__c> venues = new List<Venue__c>();
        for(Event__c ev : events){
            if(ev.Venue__c != null){
                if(ev.Event_Status__c == 'Confirmed'){
                    venues.add(new Venue__c(Id=ev.Venue__c, Availability__c='Reserved'));
                } else if(ev.Event_Status__c == 'Canceled'){
                    venues.add(new Venue__c(Id=ev.Venue__c, Availability__c='Available'));
                }
            }
        }
        if(!venues.isEmpty()) update venues;
    }
}

```

EventTrigger13.trigger

```
trigger EventTrigger13 on Event__c (after insert, after update) {
    VenueStatusHelper.updateVenueStatus(Trigger.new);
}
```

PreventDoubleBooking.trigger

```
trigger PreventDoubleBooking on Event__c (before insert, before update) {
    Set<Id> venueIds = new Set<Id>();
    for(Event__c ev : Trigger.new){
        if(ev.Venue__c != null) venueIds.add(ev.Venue__c);
    }
    Map<Id, List<Event__c>> venueEventMap = new Map<Id, List<Event__c>>();
    for(Event__c e : [SELECT Id, Venue__c, Event_Date__c FROM Event__c WHERE Venue__c IN :venueIds]){
        if(!venueEventMap.containsKey(e.Venue__c)){
            venueEventMap.put(e.Venue__c, new List<Event__c>());
        }
        venueEventMap.get(e.Venue__c).add(e);
    }
    for(Event__c ev : Trigger.new){
        if(ev.Venue__c != null && venueEventMap.containsKey(ev.Venue__c)){
            for(Event__c existing : venueEventMap.get(ev.Venue__c)){
                if(existing.Event_Date__c == ev.Event_Date__c && existing.Id != ev.Id){
                    ev.addError('This Venue is already booked on this date.');
```

BatchCompleteEvents.cls

```
global class BatchCompleteEvents implements Database.Batchable<sObject> {
    global Database.QueryLocator start(Database.BatchableContext bc){
        return Database.getQueryLocator(
            'SELECT Id, Event_Date__c, Event_Status__c FROM Event__c ' +
            'WHERE Event_Date__c < TODAY AND Event_Status__c != \'Completed\'' );
    }
    global void execute(Database.BatchableContext bc, List<Event__c> scope){
        for(Event__c ev : scope){ ev.Event_Status__c = 'Completed'; }
        update scope;
    }
    global void finish(Database.BatchableContext bc){
        System.debug('Past events marked as Completed');
    }
}
```

ScheduleCompleteEvents.cls

```
global class ScheduleCompleteEvents implements Schedulable {
    global void execute(SchedulableContext sc){
        Database.executeBatch(new BatchCompleteEvents(), 200);
    }
}
```

Scheduling the batch job

From Setup > Apex Classes > **Schedule Apex**: job name **Daily Event Completion**, class

ScheduleCompleteEvents, frequency Weekly with all days selected, preferred start time **8:00 PM**. Salesforce then marks past events as Completed every evening.

Apex Classes

Apex Code is an object oriented programming language that allows developers to develop on-demand business applications on the Lightning Platform.

Percent of Apex Used: 0.02%
You are currently using 362 characters of Apex Code (excluding comments and @test annotated classes) in your organization, out of an allowed limit of 6,000,000 characters. Note that the amount in use includes both Apex Classes and Triggers defined in your organization.

Estimate your organization's code coverage |
Compile all classes |

View: [All] Create New View

Action	Name	Namespace Prefix	API Version	Status	Size Without Comments	Last Modified By	Has Trace Flags
BatchCompleteEvents	BatchCompleteEvents	devodapp	67.0	Active	661	Naureen P 10/2/2026, 7:39 PM	☐
DeveloperEditionUtils	DeveloperEditionUtils	devodapp	64.0	Active	164	OrgEam.EPIC, 9/26/2025, 9:23 AM	☐
PostInstallScript	PostInstallScript	devodapp	64.0	Active	2,175	OrgEam.EPIC, 9/26/2025, 9:23 AM	☐
ScheduleCompleteEvents	ScheduleCompleteEvents	devodapp	64.0	Active	761	OrgEam.EPIC, 9/26/2025, 9:23 AM	☐
VenueStatusHelper	VenueStatusHelper	devodapp	67.0	Active	160	Naureen P 10/2/2026, 7:40 PM	☐

Figure 12. Apex Classes list in Setup: BatchCompleteEvents, ScheduleCompleteEvents and VenueStatusHelper (API 67.0, Active).

04 Phase 3: UI/UX Development and Customization

Tabs, the Event Planner app, reports and dashboard

Milestone 9: Custom tabs

A custom tab was created for each object (Setup > Tabs > Custom Object Tabs > New) so that users can reach records quickly: **Clients, Events, Feedbacks, Vendors and Venues**.

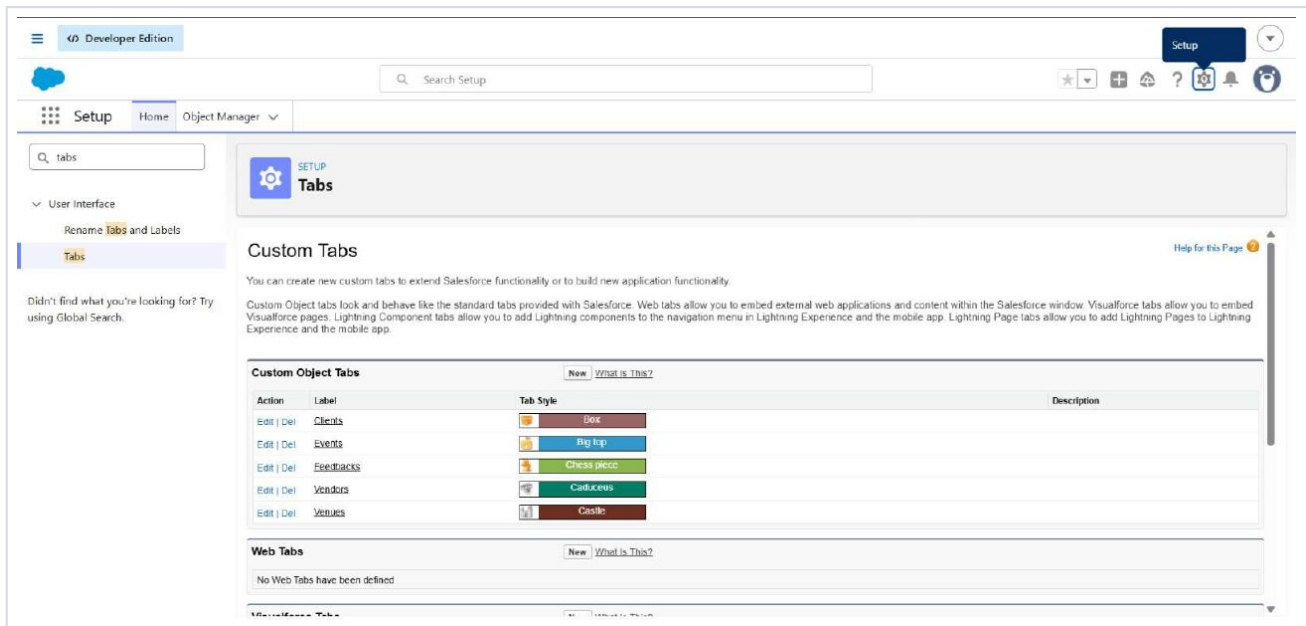


Figure 13. Custom Object Tabs in Setup.

Milestone 10: Lightning App

The Lightning App **Event Planner** groups the project in one place (Setup > App Manager > New Lightning App). Its navigation bar contains Events, Clients, Vendors, Venues, Feedbacks, Reports and Dashboards, so a user does not need to search Setup to do daily work. The app was made available to the required profiles.

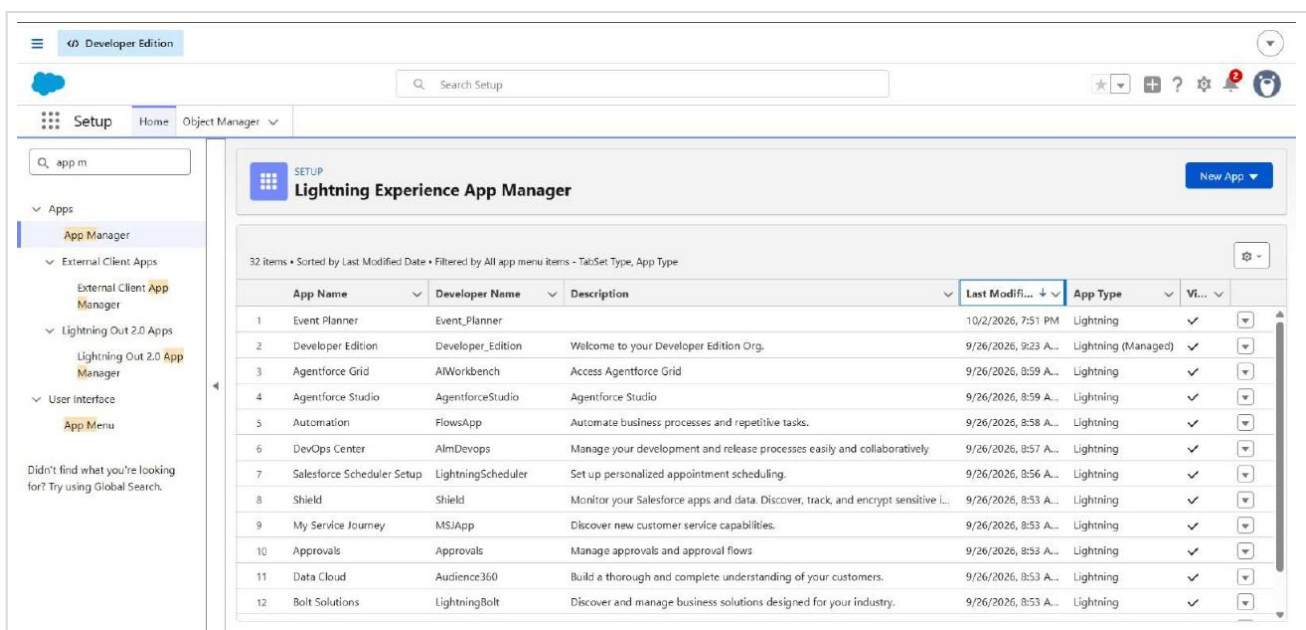


Figure 14. Lightning Experience App Manager showing the Event Planner app.

Milestone 11: Reports

A report folder named **EventForce Report** was created and shared by role (Event Coordinator and Vendor Manager with View access). Records were added to every object first so that the reports have data to show.

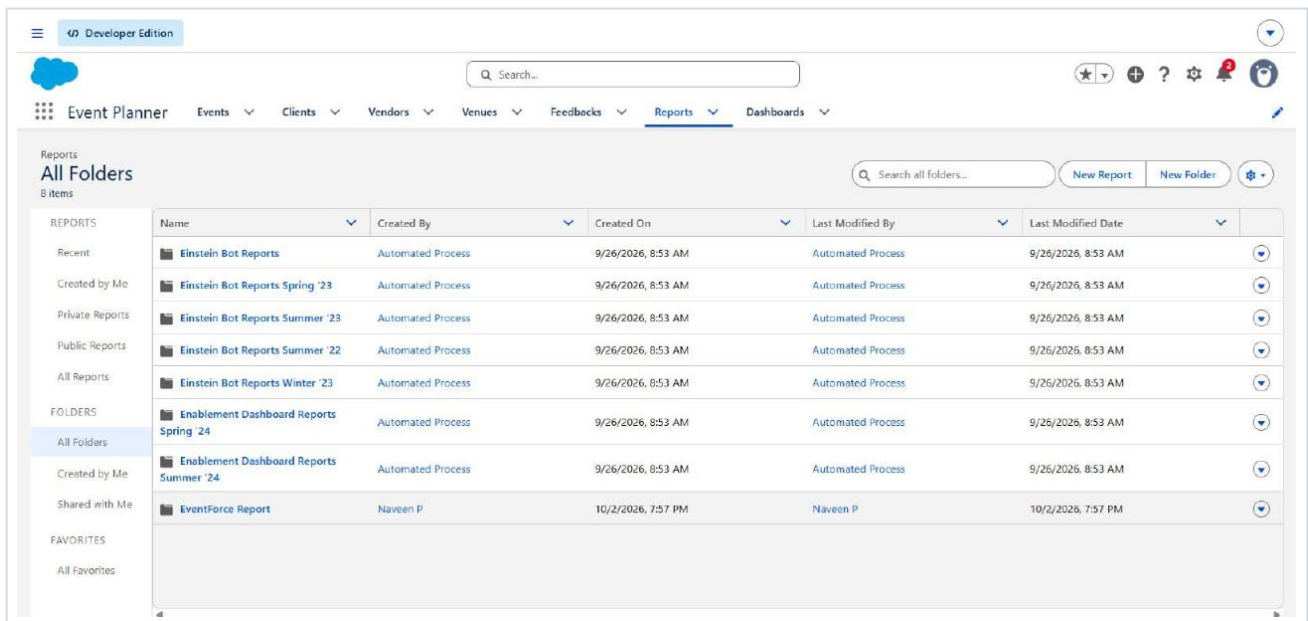


Figure 15. Reports tab in the Event Planner app with the EventForce Report folder.

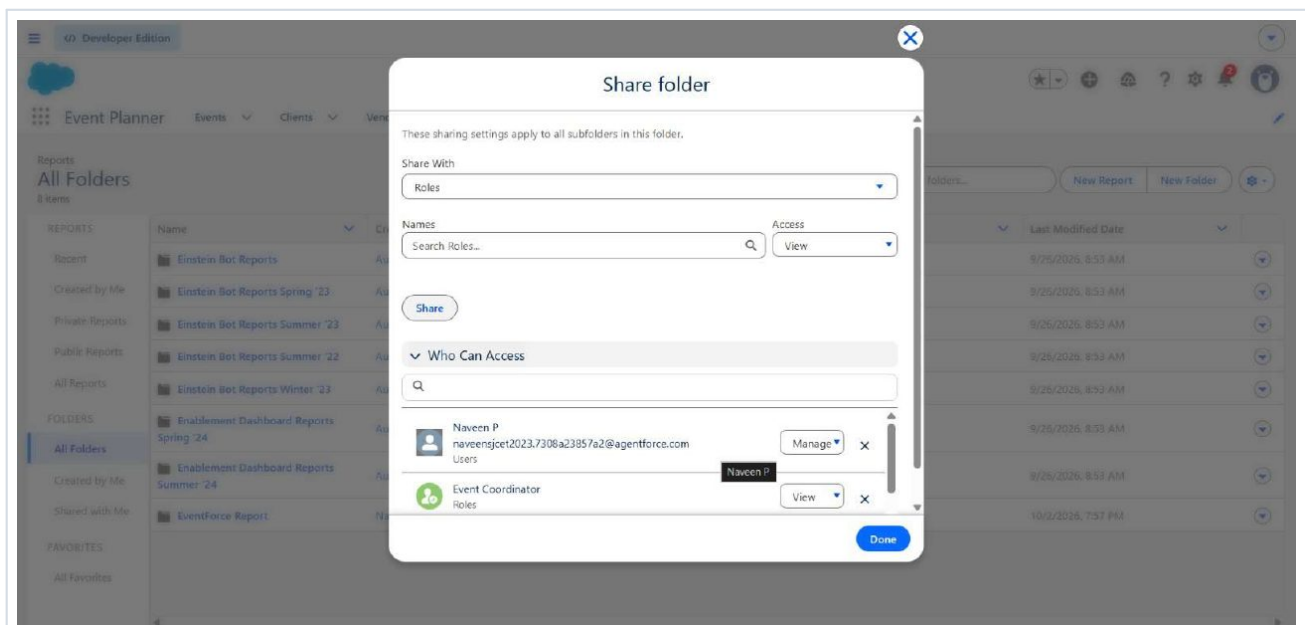


Figure 16. Sharing the report folder with roles (Event Coordinator: View).

Report 1: New Events with Venue Report. Uses the Events with Venue report type and shows Event Name, Event Date, Venue Name, Event Type, Client and Event Status for each event.

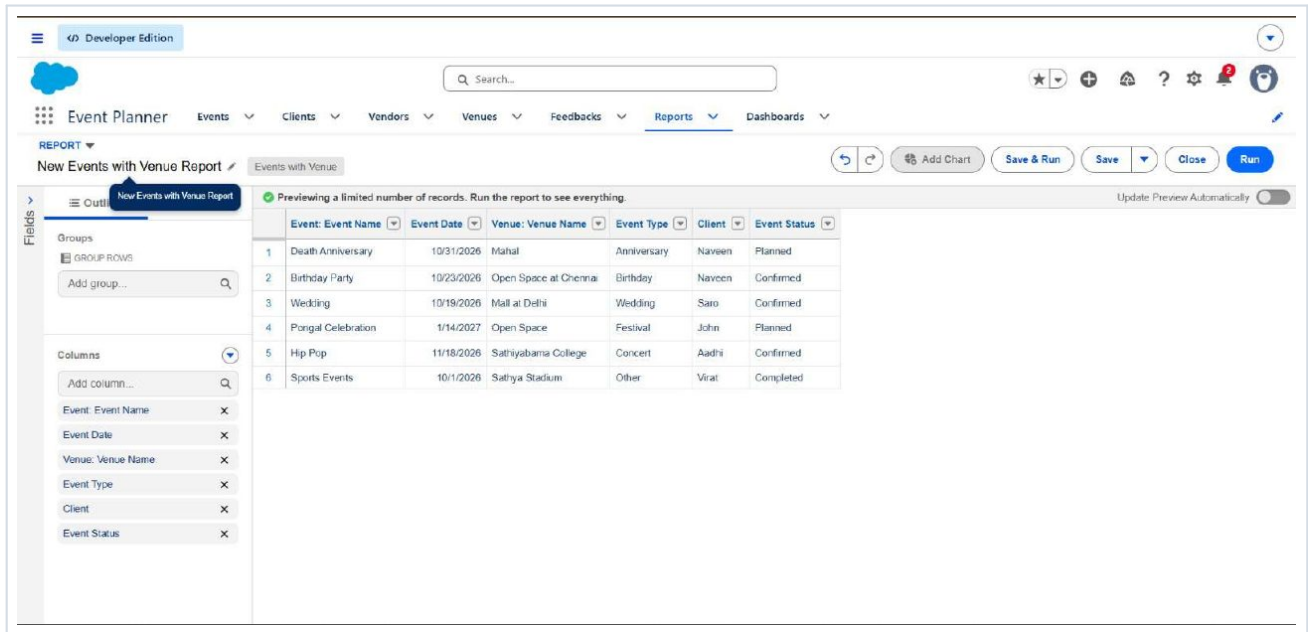


Figure 17. New Events with Venue Report (report builder).

Report 2: Upcoming Events by Month. Rows are grouped by Event Date (calendar month) with a record count and a chart, and the columns include Event Name, Venue Name, Event Budget, Event Type, Client and Event Status.

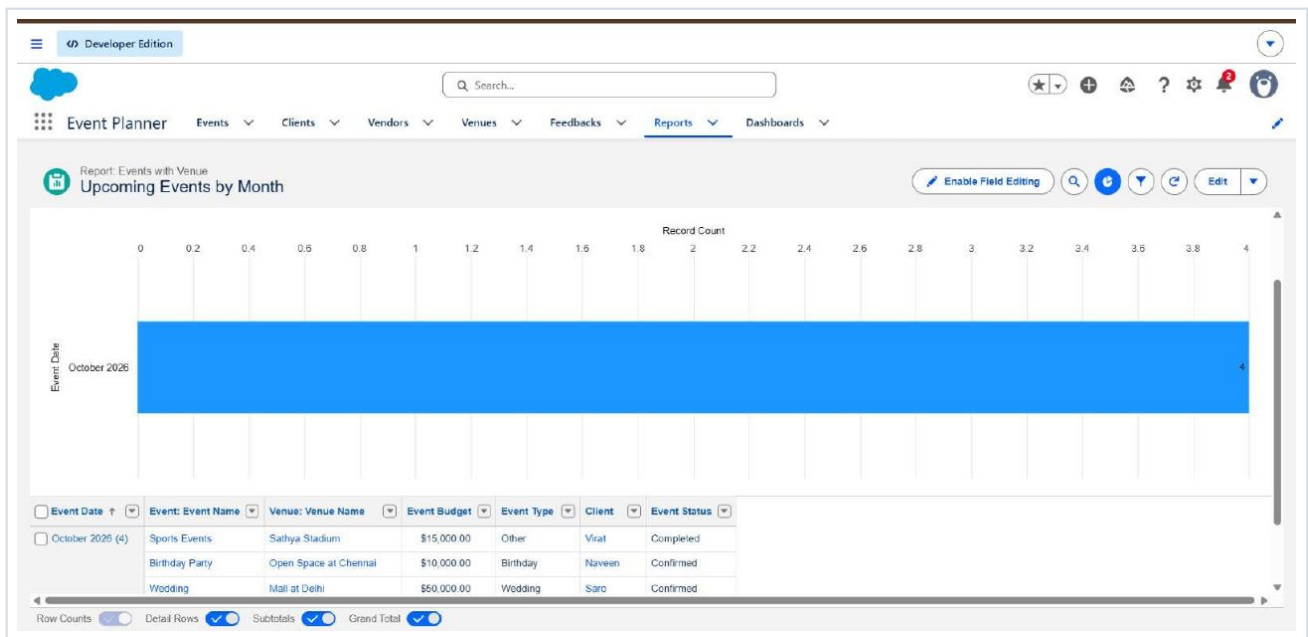


Figure 18. Upcoming Events by Month report with chart.

Milestone 12: Dashboard

The dashboard **EventForce Operations Dashboard** shows the donut chart *Upcoming Events by Month* (subtitle: Upcoming Events Overview), built from the report data. It gives coordinators a quick view of how many events are coming up.

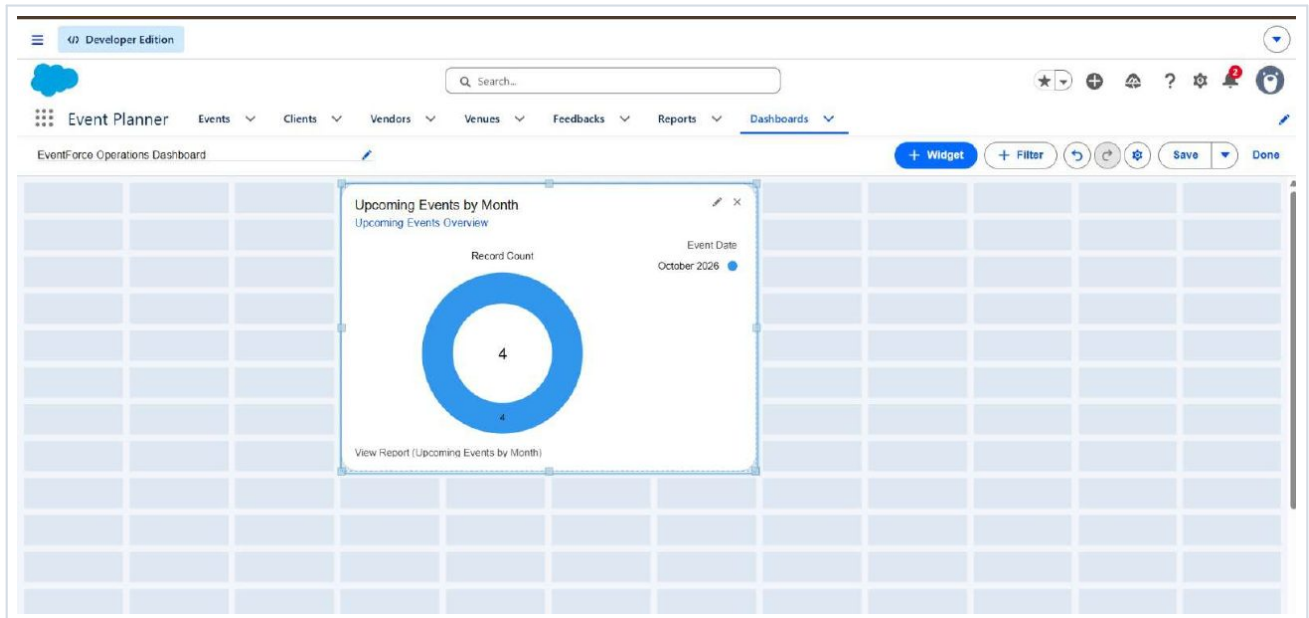


Figure 19. EventForce Operations Dashboard with the Upcoming Events by Month donut chart.

05 Phase 4: Data Migration, Testing and Security

Data loading, access control and test plan

Milestone 13: Data migration

Records for Events, Clients, Vendors, Venues and Feedback were loaded with the **Data Import Wizard** after pre-validation checks to avoid duplicates and keep the data consistent.

Milestone 14: Profiles

Custom profiles control what each user type can see and do. The profiles are **Event Admin** (cloned from System Administrator, so it inherits full access), **Event Coordinator**, **Vendor Manager** and **Client Profile** (the last three cloned from Standard Platform User with the Salesforce Platform license). Event Planner is the default app for them, the session timeout is 2 hours, passwords never expire and the minimum password length is 8.

Profile	Event	Client	Vendor	Venue	Feedback
Event Admin	Full	Full	Full	Full	Full
Event Coordinator	Create, Read, Edit, Delete	Read, Edit	Read	Read	Create, Read
Vendor Manager	Read	Read	Create, Read, Edit, Delete	Read	Read
Client Profile	Read (own events)	-	No access	No access	Create, Read

The screenshot shows the Salesforce Setup interface for the 'Client Profile'. The left sidebar contains navigation links for 'Setup', 'Home', and 'Object Manager'. The main content area is titled 'Profiles' and shows the 'Client Profile' details. The profile is a custom profile for the 'Salesforce Platform' license. The 'Profile Detail' section shows the name 'Client Profile', user license 'Salesforce Platform', and creation/modification dates. The 'Page Layouts' section shows the standard object layouts for various objects, including Global, Email Application, Home Page Layout, Account, Lead, Location, Location Group, and Location Group Assignment.

Figure 20. Client Profile detail (custom profile, Salesforce Platform license).

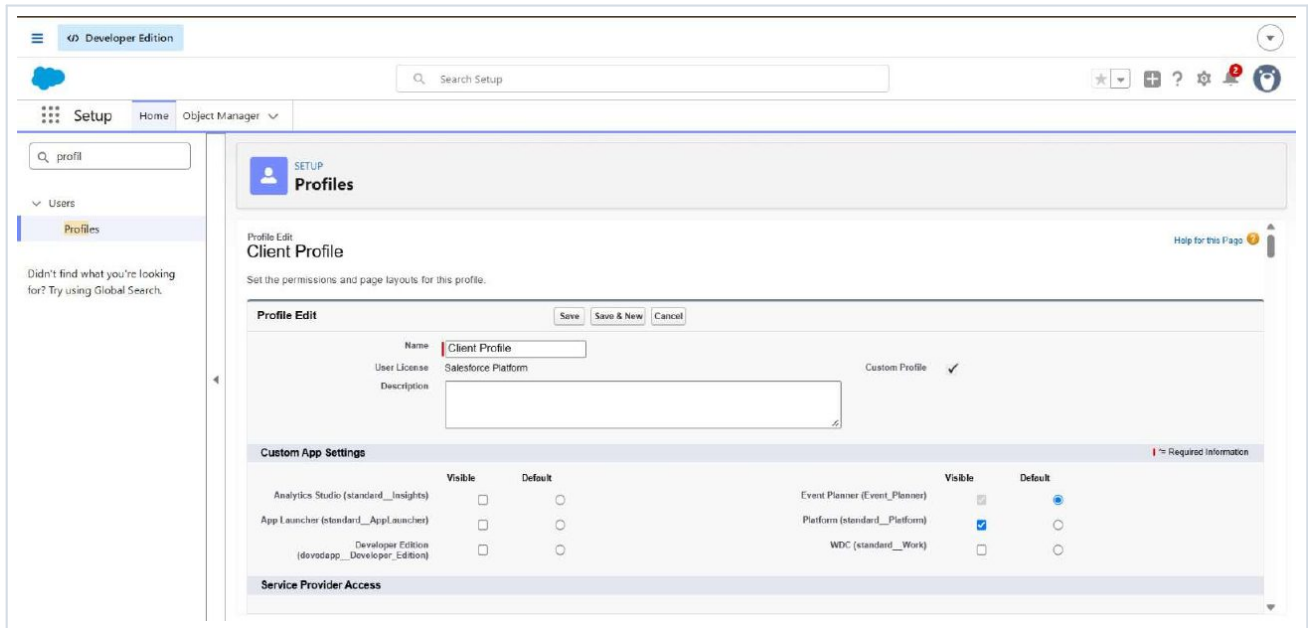


Figure 21. Client Profile edit: Event Planner set as the default custom app.

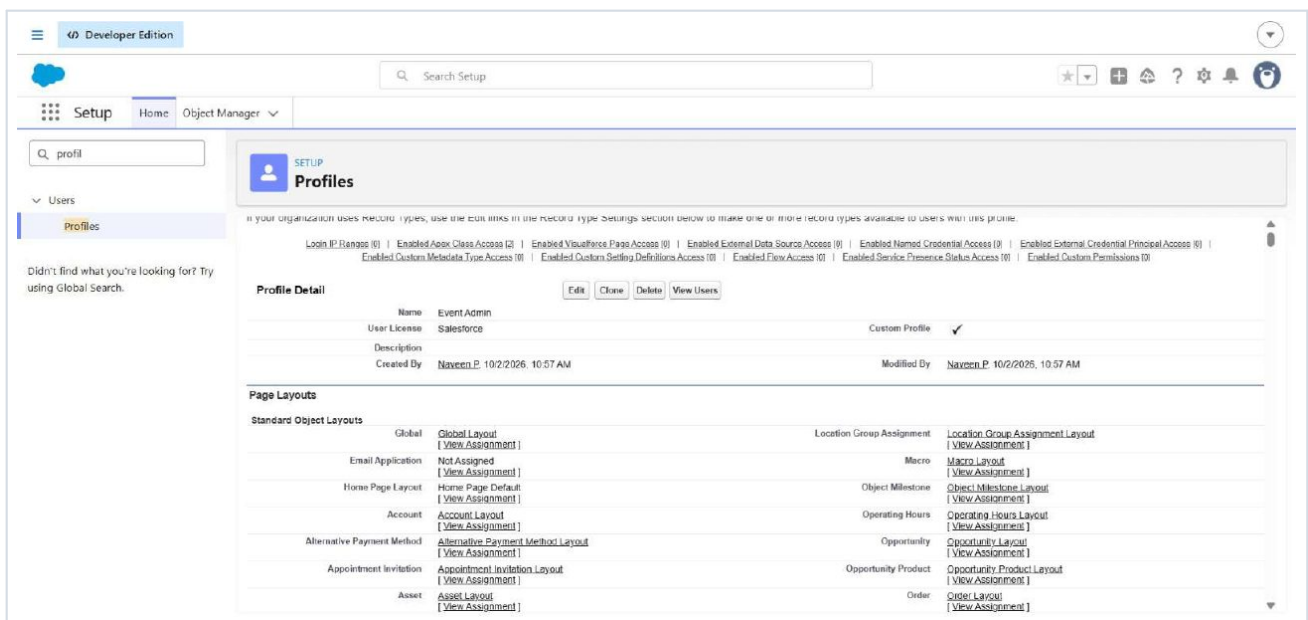


Figure 22. Event Admin profile (custom profile, Salesforce license).

Milestone 15: Roles

The role hierarchy places **Event Admin** under the CEO. The roles **Client**, **Event Coordinator** and **Vendor Manager** sit below Event Admin, so records owned by them are visible upward in the hierarchy.

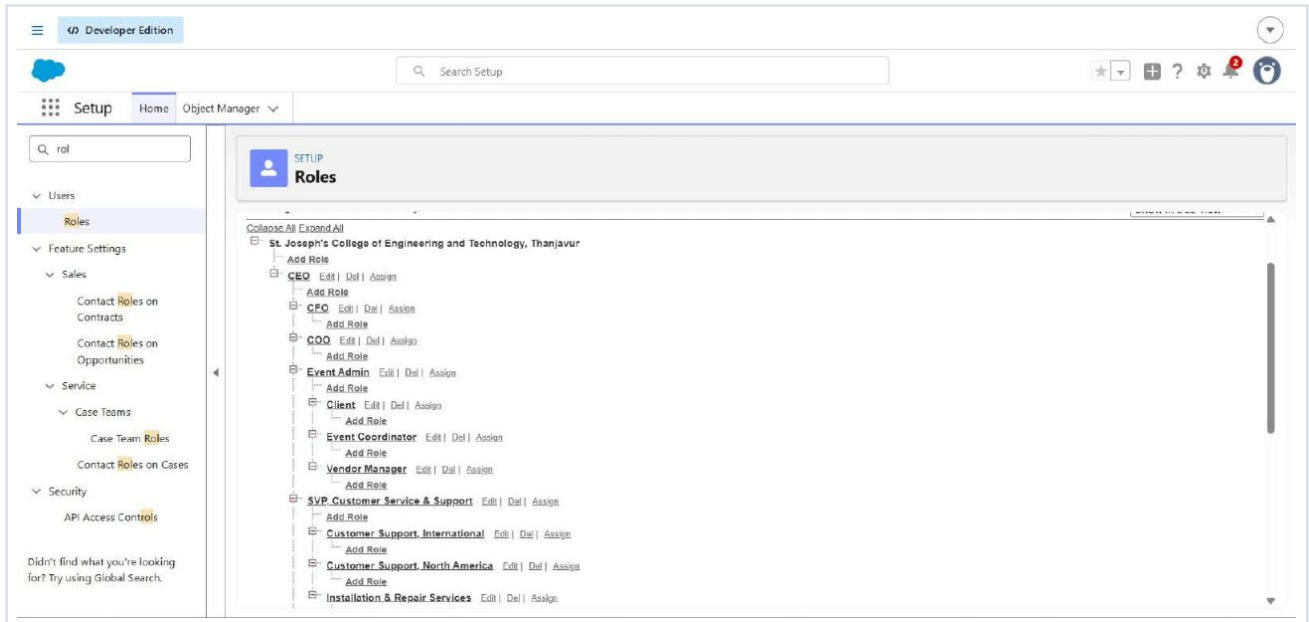


Figure 23. Role hierarchy with Event Admin and the three roles below it.

Milestone 16: Users

User	Role	Profile	License
Sizzler Jaligam	Event Admin	Event Admin	Salesforce
Roopan G	Event Coordinator	Event Coordinator	Salesforce Platform
Saravanan S	Vendor Manager	Vendor Manager	Salesforce Platform

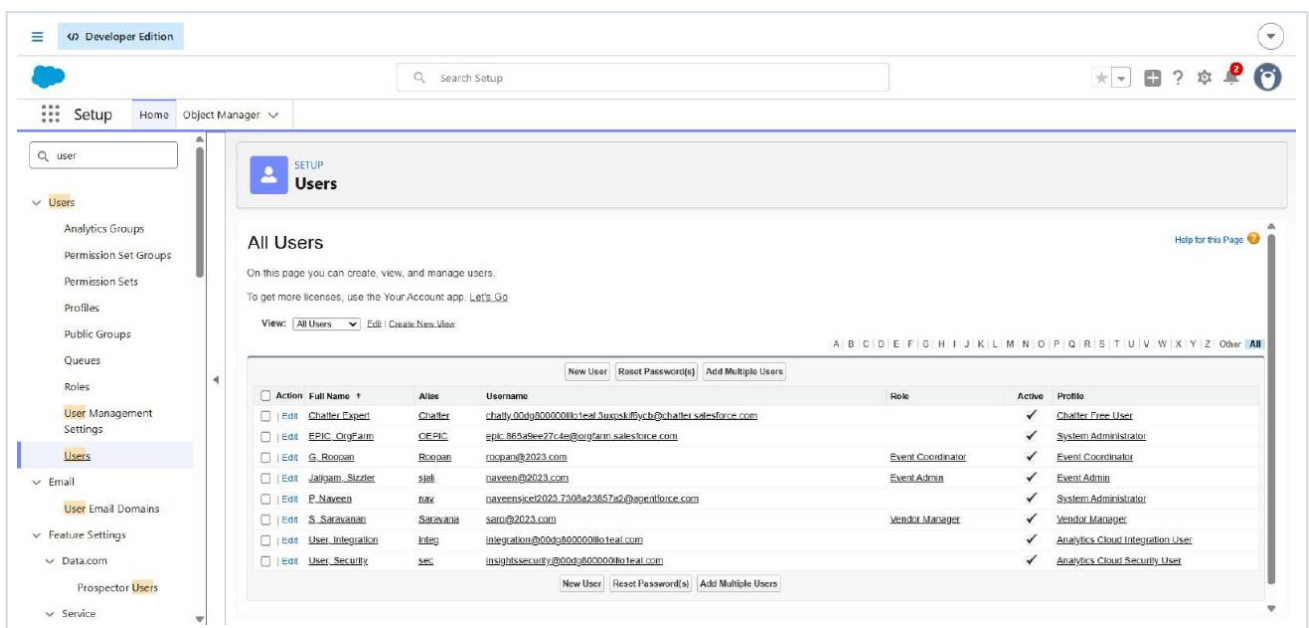


Figure 24. All Users list.

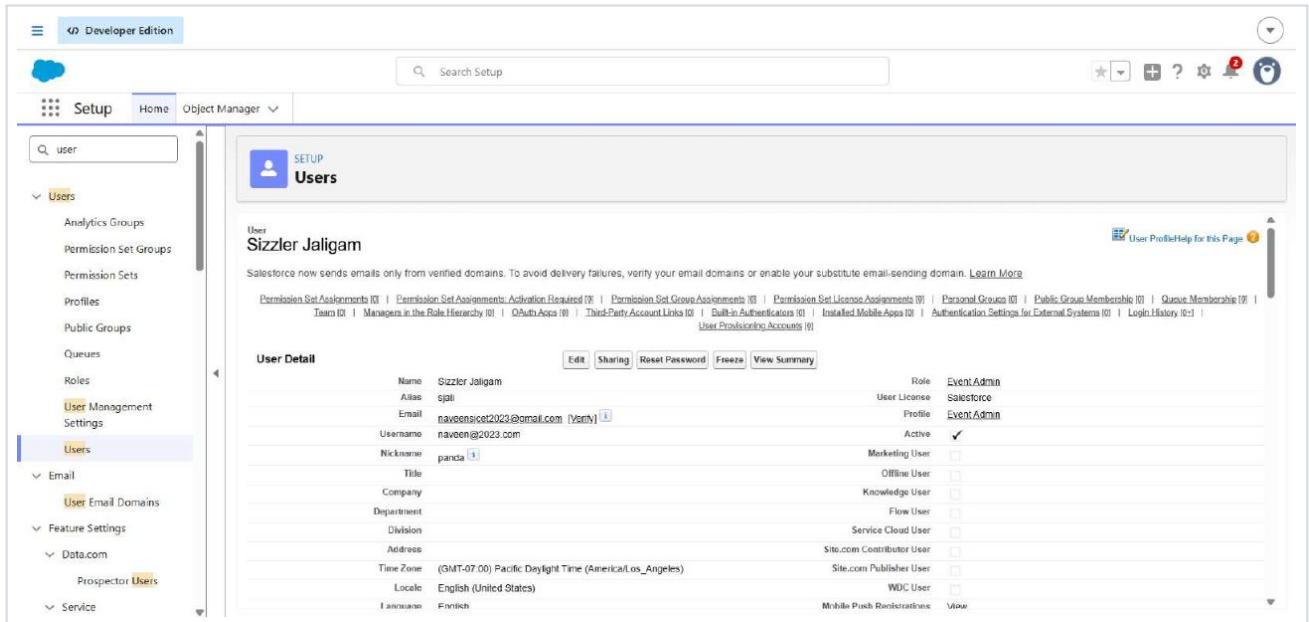


Figure 25. User detail: Sizzler Jaligam (Event Admin role and profile).

Milestone 17: Permission set

The permission set **Feedback Manager** adds extra access on top of a profile without changing the profile itself. In Object Settings, the Feedback object was given Read, Create, Edit and Delete. The set is assigned to the user **Roopan G** (Event Coordinator).

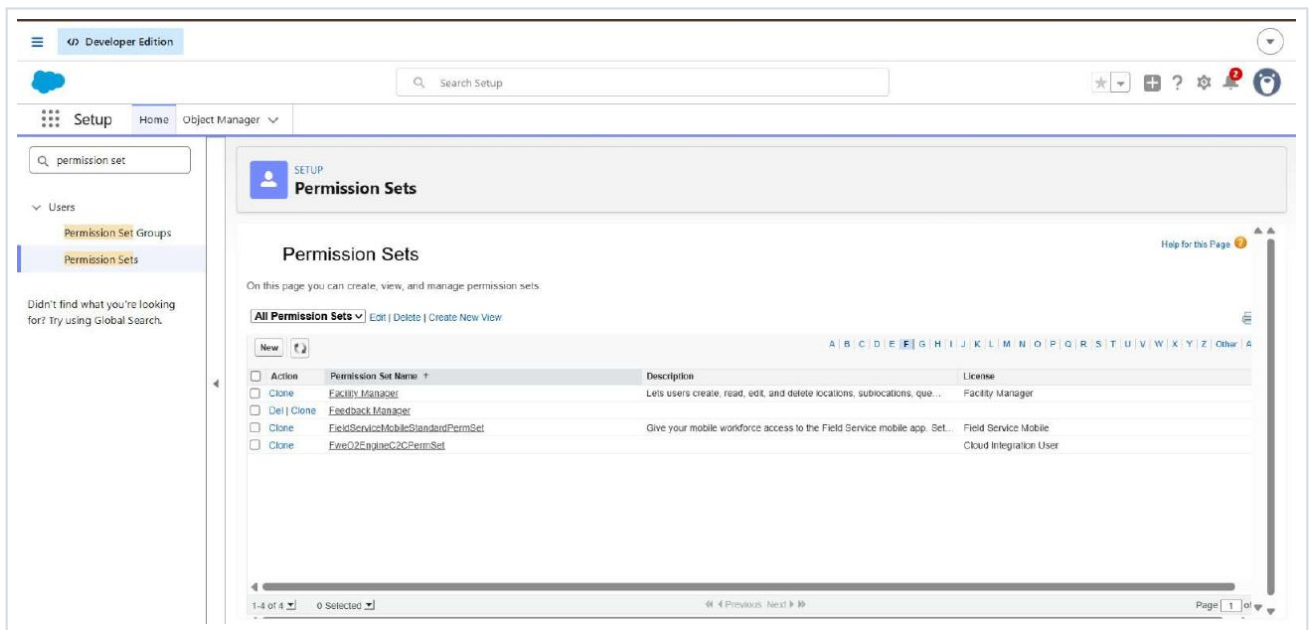


Figure 26. Permission Sets list showing Feedback Manager.

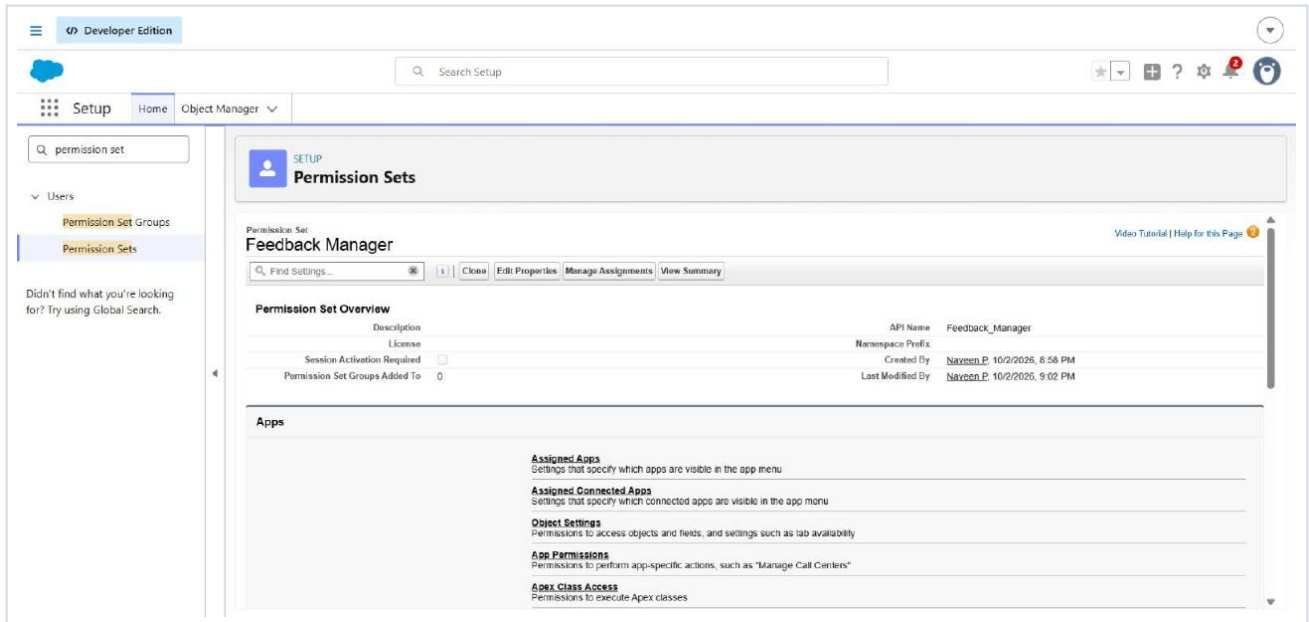


Figure 27. Feedback Manager permission set overview.

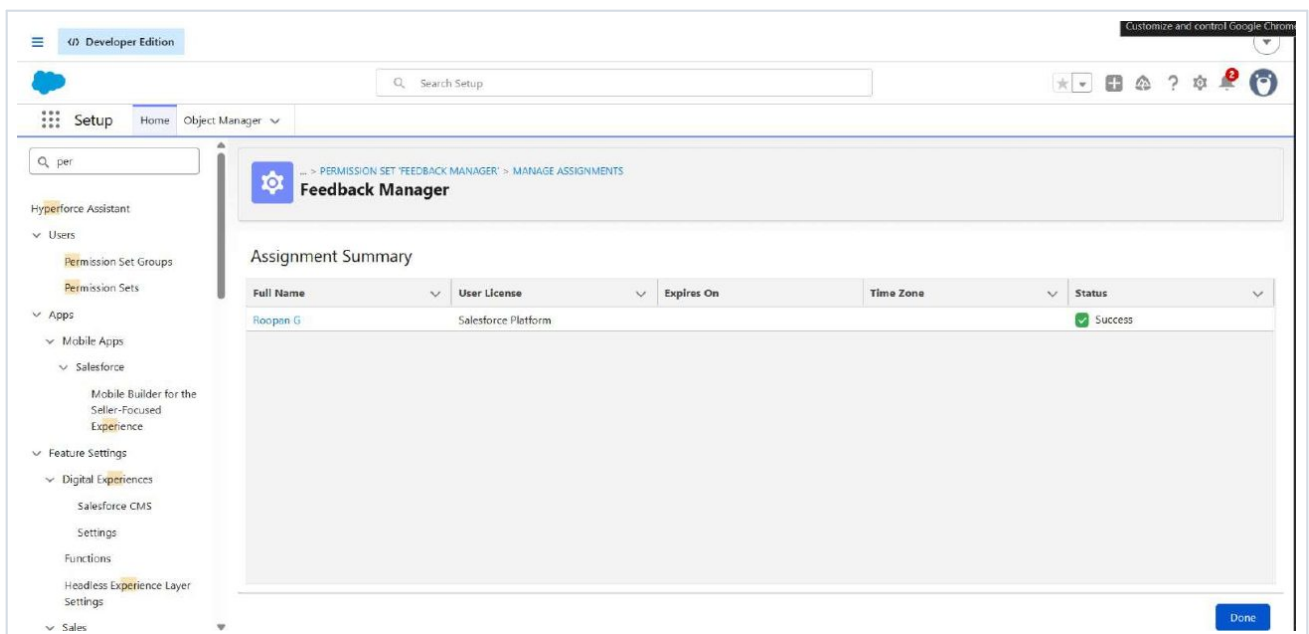


Figure 28. Assignment summary: Feedback Manager assigned to Roopan G.

Milestone 18: Sharing settings

Organization-Wide Defaults set the baseline record access. **Event is Private**, so users see only their own events and those of users below them in the role hierarchy. **Client, Feedback, Vendor and Venue are Public Read/Write** because these are shared reference records. "Grant Access Using Hierarchies" is enabled for the custom objects.

Object	Default internal access	Default external access
Event	Private	Private
Client	Public Read/Write	Private
Feedback	Public Read/Write	Private
Vendor	Public Read/Write	Private
Venue	Public Read/Write	Private

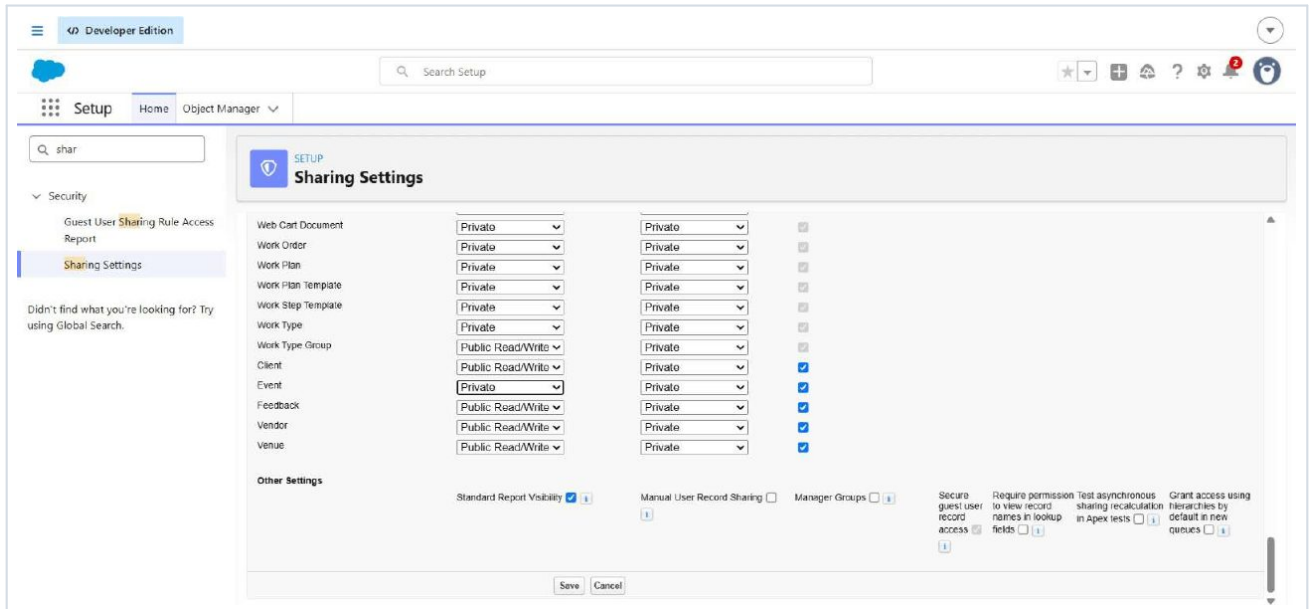


Figure 29. Organization-Wide Defaults for the custom objects (Event: Private).

Sharing rule. Because Event is Private, a criteria/owner-based rule **Event_Sharing_For_Vendors** shares events owned by the **Event Coordinator** role with the **Vendor Manager** role at **Read Only** access, so vendor managers can see the events they need to supply.

Test plan

The table lists the checks planned for the main features. The **Result** column is ticked after each test is run in the org, with a screenshot kept as evidence.

ID	Feature	Test steps	Expected result	Result
TC-01	Email validation	Create a Client with an invalid email such as "abc@".	Save is blocked with "Please Enter Valid Email Address".	[] Pass
TC-02	Double booking	Create two Events at the same Venue on the same date.	Second save fails with "This Venue is already booked on this date."	[] Pass
TC-03	Venue reserved	Set an Event to Confirmed.	Venue Availability Status becomes Reserved.	[] Pass
TC-04	Venue available	Set the Event to Canceled.	Venue Availability Status becomes Available.	[] Pass
TC-05	Cancellation approval	Set Event Status to Pending Cancellation, submit for approval and approve it.	Record locks on submit; on approval the status becomes Canceled and an email alert is sent.	[] Pass
TC-06	Cancellation rejection	Repeat TC-05 and reject the request.	Record unlocks, status changes to Rejected, rejection email is sent.	[] Pass
TC-07	Client reminder Flow	Create a Confirmed Event dated a few days ahead.	Scheduled path sends the reminder 3 days before the Event Date.	[] Pass
TC-08	Batch job	Run BatchCompleteEvents with a past, non-completed Event.	The past Event is set to Completed.	[] Pass
TC-09	Access control	Log in as Event Coordinator and Vendor Manager.	Each user sees only what the profile, role and OWD allow.	[] Pass

ID	Feature	Test steps	Expected result	Result
TC-10	Sharing rule	As Vendor Manager, open an Event owned by an Event Coordinator.	The Event is visible with Read Only access.	[] Pass

06 Plan and Schedule

Agile sprints and story points

We followed an Agile approach: work was split into epics and user stories, each sized in story points, and the team velocity was used to check that we stayed on track. The work was done in four sprints.

Sprint	Epic	Story	User story / task	Pts	Priority	Owner
1	Event Management	US-01	As an Event Coordinator, I want to create events with type, date and status so that event information is standardized.	3	High	Naveen P
1	Client and Venue Management	US-02	As an Event Coordinator, I want to associate a client and venue with an event so that coordination is centralized.	5	High	Saravanan K
2	Vendor Management	US-03	As a Vendor Manager, I want to assign multiple vendors to an event so that services can be coordinated.	5	High	Roopan G
2	Notifications and Automation	US-04	As a Client, I want timely event reminders so that I know when my event is approaching.	3	High	Mukil Doss S
3	Approval Management	US-05	As an Event Coordinator, I want cancellation requests to follow an approval process so that cancellations are controlled.	5	High	Naveen P
3	Venue Management and Apex	US-06	As a Venue Manager, I want double bookings prevented so that venue schedules remain accurate.	8	High	Saravanan K
4	Security and Access Control	US-07	As an Event Administrator, I want role-based access so that users see only appropriate information.	5	High	Roopan G
4	Reports and Dashboards	US-08	As Management, I want reports and dashboards so that event operations can be monitored.	5	Medium	Mukil Doss S

Sprint summary and velocity

Sprint	Planned points	Completed points	Stories
Sprint 1	8	8	US-01, US-02
Sprint 2	8	8	US-03, US-04
Sprint 3	13	13	US-05, US-06
Sprint 4	10	10	US-07, US-08
Total	39	39	8 user stories

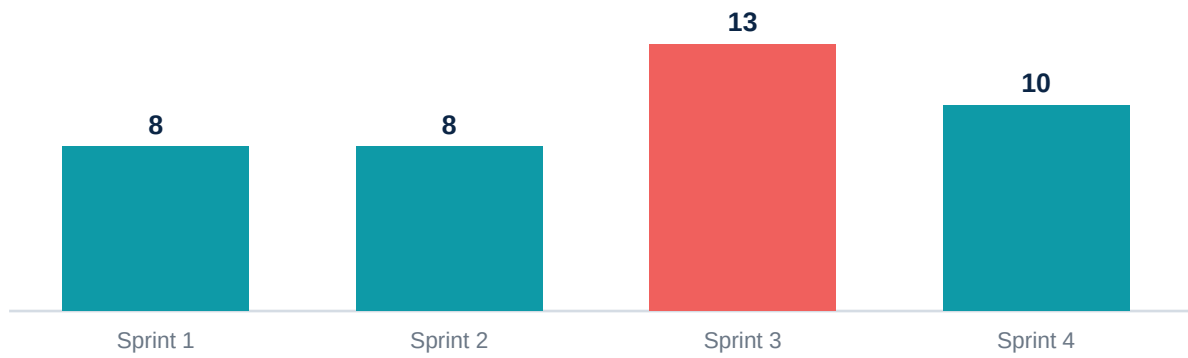


Figure C. Story points per sprint (total 39, peak load in Sprint 3 shown in coral).

Velocity. 39 story points over 4 sprints is about 9.8 points per sprint. Sprint 3 carried the most work because the Apex double-booking logic and the approval process were the largest parts of the project. The build in the org was carried out on 02 and 03 October 2026 (record timestamps in Salesforce).

07 Strengths and Limits

An honest look at what we built

Strengths

- All event, client, venue, vendor and feedback data sits in one connected CRM.
- Flows automate reminders, which reduces manual effort.
- A venue cannot be double booked on the same date; the error is shown before the save.
- Venue availability is updated by code with no manual step.
- Cancellations follow a manager approval with record lock and email alerts.
- Reports and a dashboard give visibility into upcoming events.
- Profiles, roles, a permission set, OWD and a sharing rule keep access under control.
- Validation rules and pre-checked data loading keep records accurate.
- A batch job marks past events as Completed.

Limits

- The double-booking check compares only the event date, not start and end times.
- Venue status follows the last saved event, so cancelling one event can show a venue as Available even if it has another confirmed event on a different date.
- The approval has a single step with one approver.
- No Apex test classes are written yet, which a production deployment would need.
- The project was built in a Developer Edition org for academic purposes and was not deployed to production.
- Real deployment needs sandboxes, Change Sets, Salesforce DX or CI/CD tools.
- Flows, Apex jobs, approvals and sharing need regular monitoring and maintenance.
- Reminder emails depend on a verified email sending domain.

08 Conclusion and Roadmap

What we learned and where we would take it next

The EventForce Management System shows how a standard Salesforce org can be shaped into a working event management system. Events, Clients, Vendors, Venues, Feedback and Event Vendor data now live on one platform. With a clear data model, the system blocks double bookings, keeps venue availability current, reminds clients before their event, routes cancellations for approval and limits access by role. Through this project the team learned object design and relationships, Flow with scheduled paths, approval processes, Batch and Schedulable Apex, data migration and the Salesforce security model. The outcome is less manual follow-up, cleaner data and a clear view of upcoming events for the whole team.

Where we would take it next

Area	Next step
Production deployment	Move the solution from the Developer org to a production org using sandboxes and proper deployment tools.
Smarter booking	Check time slots and overlapping dates, not only the event date.
Approvals	Add more approval steps and escalation when a request is not answered.
Vendor management	Add vendor tracking and performance management.
Analytics	Add reports and dashboards for vendor performance, feedback ratings and budgets.
Integrations and mobile	Connect external services (for example SMS or WhatsApp reminders) and improve the mobile experience.
Quality	Write Apex test classes with enough code coverage for a production release.
Maintenance	Regularly review Flows, Apex, validation rules and sharing settings as requirements change.

A Appendix

Reference items and team

Ref	Item	Description
A	Custom objects	Event__c, Client__c, Vendor__c, Venue__c, Feedback__c, Event Vendor (junction)
B	Validation rule	Email_Valid_Address on Client
C	Approval Process	Event_Cancellation_Process on Event
D	Flow	Client Reminder - 3 Days Before (Record-Triggered, scheduled path)
E	Apex	VenueStatusHelper, EventTrigger13, PreventDoubleBooking, BatchCompleteEvents, ScheduleCompleteEvents
F	Lightning App	Event Planner
G	Reports and Dashboard	New Events with Venue Report, Upcoming Events by Month, EventForce Operations Dashboard
H	Security	Event Admin, Event Coordinator, Vendor Manager, Client Profile; Feedback Manager permission set; OWD; Event_Sharing_For_Vendors
I	Data migration	Salesforce Data Import Wizard

Team member	Role	NM ID	Register No.
Naveen P	Team Leader	2F9886E9DAD1704907D4F8C339C7228D	821923243034
Saravanan K	Team Member	F18CF35068E4C2B82EE447E9431781E7	821923243046
Roopan G	Team Member	F85EB53C24A0D469DCC4BCD5ABF04272	821923243043
Mukil Doss S	Team Member	0F27DB0131DA011A5144D5726477679E	821923243033

St. Joseph's College of Engineering and Technology, Thanjavur | College Code: 8219 | Team ID: SWTID-2026-3509